Context-Aware Safety Evaluation and Control for Semantic Object Transport

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Abstract

Safe robot behavior is context-dependent: transporting a liquid near electronics, a knife near people, or a heavy object near fragile items should alter both the cost assigned to candidate motions and the way those motions are executed. This paper presents SARM, a semantic risk-aware planning and validation framework for safe object transport. The framework combines a CBF-inspired geometric admissibility layer, defined over both the robot body and the carried-object footprint, with pairwise semantic hazard rules encoding relation type, severity weight, and preferred soft clearance. These structured rules are grounded into continuous risk fields, allowing semantic safety to act as a graded planning signal rather than a binary blocking constraint. The same risk representation is used for trajectory generation, exposure-preserving smoothing, dynamic replanning, and dense post-hoc validation. We evaluate SARM on a completed navigation-style semantic transport benchmark with static and dynamic hazards, and describe a manipulation-style transport protocol that applies the same risk representation to Cartesian object transport. The navigation results show that the proposed framework improves contextual safety relative to geometry-only planning while avoiding the excessive conservatism and reduced feasibility of hard semantic blocking baselines.

1 Introduction

Safe robot behavior is context-dependent. A trajectory that is geometrically collision-free may still be unsafe when the robot is transporting an object whose meaning changes the risk of nearby interactions. For example, carrying a liquid near electronics, moving a knife near people, or transporting a heavy object near fragile items should affect not only how candidate trajectories are evaluated, but also how those trajectories are generated, smoothed, validated, and executed. Classical motion planning methods primarily reason about geometry: obstacles are either free or occupied, and safety is typically expressed as collision avoidance with a fixed clearance margin. While this is necessary for physical feasibility, it does not capture many task-level hazards that arise in everyday environments.

This paper presents SARM, a semantic risk-aware planning and control framework for safe object transport in structured environments. The framework augments a shared control-barrier-function-style geometric safety shield with pairwise semantic hazard rules. Each rule represents a relationship between two entities in the scene, such as liquid–electronic, sharp–human, heavy–fragile, or robot–human. The rule is parameterized by a relation type, severity weight, and preferred soft clearance. These structured relationships are then grounded into continuous risk fields over the workspace, allowing semantic safety to be treated as a graded planning signal rather than only as a hard blocking constraint.

The central design principle of SARM is to separate hard geometric feasibility from soft semantic risk. Geometric constraints, including the robot body, carried-object footprint, workspace boundaries, and physical obstacles, are enforced through a shared CBF-style admissibility layer. Semantic risk, by contrast, is represented as a continuous cost that influences path search, smoothing, dynamic replanning, and trajectory validation. This separation is important because physical collisions should remain strictly forbidden, whereas many semantic hazards are better modeled as tradeoffs. In narrow or cluttered scenes, completely blocking every semantically risky region can make the task infeasible even when a reasonable low-exposure path exists.

We evaluate SARM quantitatively in a navigation-style semantic transport benchmark where the robot carries an object through a two-dimensional workspace containing static obstacles, semantic hazards, and dynamic moving hazards. The evaluation compares SARM against geometry-only planning and conservative semantic baselines, including semantic inflation and hard risk blocking, using raw trajectory-level metrics such as semantic exposure, semantic-zone occupancy, clearance, path length, smoothness, planning time, success, and barrier-margin violations. We also describe a manipulation-style transport protocol in which the same semantic risk representation is applied to Cartesian carried-object motion before robot execution.

The main contributions of this work are as follows:

1. We introduce a semantic risk-aware planning framework for object transport that connects object-level scene semantics to executable robot motion. Unlike geometry-only planners, the framework accounts for contextual hazards induced by the carried object, such as liquid–electronic, sharp–human, and heavy–fragile interactions.

2. We propose a structured pairwise hazard representation that encodes relation type, severity weight, and preferred soft clearance, and grounds these rules into continuous workspace risk fields. This allows semantic safety to act as a graded planning signal rather than a binary blocking constraint.

3. We separate hard geometric feasibility from soft semantic exposure by combining a shared geometric admissibility layer with risk-aware search, exposure-preserving smoothing, dynamic replanning, and dense trajectory validation.

4. We evaluate the framework on navigation-style semantic transport benchmarks with static and dynamic hazards, comparing against geometry-only planning, semantic inflation, and hard risk blocking. The results show that the proposed method reduces semantic exposure while preserving feasibility in settings where conservative hard blocking can fail.

2 Related Work

The closest work to ours is the semantic safety filter framework proposed in [1], which integrates semantic scene understanding with control-barrier-function-based safety certification for robot manipulation. Their method constructs an object-level 3D semantic map from RGB-D observations, infers semantic constraints such as unsafe spatial relationships, cautious behaviors, and pose restrictions, and then enforces these constraints through a quadratic-program-based safety filter that modifies teleoperation or learned policy commands online. This work demonstrates that semantic constraints such as avoiding motion of a cup of water above a laptop, reducing motion near sensitive objects, or constraining end-effector rotation can be incorporated into formal robot safety filtering. In contrast, our work focuses on semantic risk-aware planning and evaluation rather than low-level command certification alone. We represent pairwise semantic hazards using relation type, severity weight, and soft clearance, and ground these rules into continuous risk fields that are used throughout path search, exposure-preserving smoothing, dynamic replanning, and dense trajectory validation. Thus, while prior semantic safety filtering treats semantic constraints primarily as safe-set conditions enforced during execution, our framework treats semantic safety as a graded planning signal that can trade off exposure, path efficiency, and feasibility while retaining a separate CBF-style shield for hard geometric safety. This distinction allows our method to handle navigation and manipulation-style transport scenarios, including dynamic hazards and narrow semantic bottlenecks where hard blocking may be overly conservative.

Recent work has begun to study responsible robotic manipulation at the level of task reasoning, hazard recognition, and benchmark evaluation. Safety-as-Policy [2] introduces a framework for developing safety-aware manipulation behavior through generated risky scenarios and iterative safety cognition, together with the SafeBox dataset covering electrical, fire/chemical, and human-related hazards. ResponsibleRobotBench [3] complements this direction by providing a benchmark suite for evaluating robot agents on multi-stage manipulation tasks with electrical, fire/chemical, and human hazards, multiple action representations, and metrics such as success rate, safety rate, safe success rate, hazard detection, and execution cost. These works demonstrate the importance of contextual hazard recognition in manipulation tasks. However, they mainly address high-level task-policy generation, agent evaluation, and benchmarking. In contrast, our framework focuses on the motion-planning and trajectory-evaluation layer: semantic hazards are represented as pairwise weighted rules with soft clearances, grounded into continuous risk fields, and used directly for path search, exposure-preserving smoothing, dynamic replanning, and dense trajectory validation. As a result, semantic safety influences the geometry and execution of the trajectory itself, rather than only the selection of evaluation of a task-level action sequence.

3 Problem Formulation

We consider a robot transporting an object through a structured environment containing physical obstacles and semantic hazards. The workspace is denoted by $\mathcal{W} \subset R^2$ for navigation-style transport and by a Cartesian task space for manipulation-style transport. The scene contains a set of objects

$$\mathcal{O} = \{o_1, \ldots, o_N\}, \tag{1}$$

where each object o_i has a position p_i, radius ρ_i, semantic category c_i, and optional tags such as liquid, electronic, human, sharp, fragile, or heavy.

The robot state is denoted by q, and the transported object is represented by a carried-object footprint attached to the robot state. A trajectory is a continuous curve

$$\tau = \{q(t) : t \in [0, T]\}, \tag{2}$$

connecting a start state q_{start} to a goal state q_{goal}.

The objective is to find a feasible trajectory that minimizes both geometric motion cost and semantic exposure:

$$J(\tau) = \int_0^T \left[\lambda_\ell \|\dot{q}(t)\| + \lambda_r r(q(t), t)\right] dt, \tag{3}$$

subject to collision-avoidance, workspace-boundary, and dynamic-feasibility constraints. Here $r(q(t), t)$ is the semantic risk field induced by pairwise hazard rules between the carried object, robot body, and nearby scene objects.

We separate hard and soft safety requirements. Hard geometric constraints define the feasible set $\mathcal{Q}_{\text{safe}}$ and must not be violated. Semantic hazards define a continuous risk cost that should be reduced when possible, but are not automatically converted into hard obstacles.

4 Method

4.1 Overview

The proposed method consists of five connected components: semantic hazard construction, continuous risk field generation, geometic shielding, risk-aware planning, and dense trajectory validation. The input is a structured scene description containing metadata on object names, categories, tags, positions, and radii. The output is a feasible path or trajectory that transports the robot and the carried object from start to goal while reducing contextual semantic risk.

The framework is designed so that semantic reasoning and geometric safety are coupled but not collapsed into the same representation. Semantic relationships determine how costly it is to move through different parts of the workspace. Geometric constraints determine whether a state or transition is physically allowed. This separation allows the planner to remain robust in a cluttered scene: it can avoid semantic exposure when possible, but it does not unnecessarily convert every soft hazard into an impassable obstacle.

4.2 Semantic Hazard Rules

A semantic hazard rule is represented as

$$h_{ij} = (c_i, c_j, \alpha_{ij}, w_{ij}, d_{ij}^{\text{soft}}), \tag{4}$$

where c_i and c_j are semantic categories, α_{ij} is the relation type, w_{ij} is the severity weight, and d_{ij}^{soft} is the preferred soft clearance.

4.3 Risk-Field Construction

For each active rule, we compute a distance-dependent risk contribution

$$r_{ij}(q) = w_{ij} \exp\left(-\frac{(d_{ij}(q))^2}{2\sigma_{ij}^2}\right), \tag{5}$$

where $d_{ij}(q)$ is the clearance between the transported object or robot body and the semantic hazard object. The total semantic risk is the sum of active rule contributions:

$$r(q) = \sum_{(i,j)\in\mathcal{H}} r_{ij}(q). \tag{6}$$

4.4 Geometric CBF Shield

4.5 Risk-Aware Search and Smoothing

4.6 Dynamic Replanning and Local Control

4.7 Manipulation-Style Transport

5 Experiments and Results

The experimental evaluation is organized into two parts. The first part is a completed navigation-style semantic transport benchmark, where the robot transports an object through two-dimensional workspaces containing static obstacles, semantic hazards, and dynamic moving hazards. This

Table 1: Overall raw navigation benchmark results. Values are mean ± standard deviation over successful trials, while the success column reports successful trials out of all 81 trials. Lower is better for all metrics except success. No balanced score is used.

Method	Success	L (m)	T (s)	E_d	E_t	Z_d (m)	Z_t (s)	Turn/m	Plan (ms)	CBF viol.
Geom-A*	81/81	11.477 ± 1.823	19.129 ± 3.038	1.197 ± 0.676	1.987 ± 1.079	2.942 ± 1.857	4.786 ± 2.883	0.113 ± 0.071	120.0 ± 39.3	0.062
Hard Risk Block	49/81	12.452 ± 2.539	22.888 ± 6.973	0.437 ± 0.394	0.831 ± 0.766	0.952 ± 1.286	1.578 ± 2.132	0.270 ± 0.767	247.0 ± 187.6	0.082
SARM	81/81	12.232 ± 2.480	22.340 ± 7.490	0.423 ± 0.497	0.781 ± 0.998	0.640 ± 1.313	1.217 ± 2.797	0.204 ± 0.153	242.8 ± 196.2	0.111
Semantic Inflation	73/81	12.012 ± 1.868	20.020 ± 3.114	0.708 ± 0.547	1.240 ± 0.941	1.325 ± 1.448	2.401 ± 2.641	0.130 ± 0.085	123.5 ± 42.0	0.068

benchmark evaluates whether semantic risk changes the path selected by the planner while preserving geometric feasibility. The second part is a robot-arm manipulation-style transport study, where the same semantic risk representation is applied to Cartesian transport motions before robot execution. Since the manipulation experiments are still being finalized, we report the navigation benchmark quantitatively and keep the manipulation section as a protocol-level description.

5.1 Navigation-Style Semantic Transport

Benchmark setup. The navigation benchmark contains 27 scene specifications with three randomized variants per scene, giving 81 trials for each method. The scenes include four demonstration scenes, 14 easier semantic transport scenes, three hard static bottleneck scenes, and six hard dynamic scenes. The semantic relations include liquid–electronics, sharp–human, heavy–fragile, and robot–human hazards. Each trial requires the robot to move from a start to a goal while accounting for both the robot body and the carried-object footprint.

We compare four methods. *Geom-A** is a geometry-only A* planner using the shared geometric safety shield but no semantic cost. *Semantic Inflation* inflates obstacles according to semantic categories and then plans geometrically. *Hard Risk Block* treats semantic risk zones as forbidden regions. SARM is the proposed soft semantic-risk-aware planner with risk-aware smoothing, dynamic replanning, and the same geometric shield.

Raw metrics. We intentionally do not use a combined balanced score in this section. Instead, we report raw validation quantities. For a path parameterized by distance s and time t, distance-integrated semantic exposure is

$$E_d = \int_0^L r(q(s))\, ds, \tag{7}$$

and time-integrated semantic exposure is

$$E_t = \int_0^T r(q(t), t)\, dt, \tag{8}$$

where $r(\cdot)$ is the semantic risk field evaluated at the carried-object/robot state. We also report semantic-zone occupancy distance Z_d, semantic-zone occupancy time Z_t, path length L, travel time T, turning per meter, planning time, success count, and dense CBF-margin violations. Non-success metrics are averaged over successful trials only; failures are represented separately by the success count.

Navigation results. The geometry-only planner succeeds in all trials and produces the shortest paths, but it also produces the largest semantic exposure: $E_d = 1.197$ and $E_t = 1.987$ on average. In contrast, SARM also succeeds in all 81 trials while reducing distance-integrated exposure to $E_d = 0.423$ and time-integrated exposure to $E_t = 0.781$. This corresponds to a 64.7% reduction in distance exposure and a 60.7% reduction in time exposure relative to Geom-A*. SARM also reduces semantic-zone occupancy from $Z_d = 2.942$ m and $Z_t = 4.786$ s to $Z_d = 0.640$ m and $Z_t = 1.217$ s.

Compared with Semantic Inflation, SARM has higher success, 81/81 versus 73/81, and lower overall raw exposure, $E_d = 0.423$ versus 0.708 and $E_t = 0.781$ versus 1.240. Hard Risk Block achieves low exposure on the subset of trials it solves, but it fails 32 of 81 trials, including 14 of the 18 hard dynamic trials and 7 of the 9 hard static trials. These results show why semantic hazards should not be converted directly into hard obstacles: hard blocking can avoid risk when a safe passage remains, but it becomes overly conservative in narrow or dynamic settings. SARM instead maintains the feasibility of soft-risk tradeoffs while still reducing semantic exposure relative to geometry-only planning.

The cost of this behavior is additional path and computation overhead. Relative to Geom-A*, SARM increases mean path length from 11.477 m to 12.232 m and mean planning time from 120.0 ms to 242.8 ms. This overhead is expected because the planner searches over the same geometric domain while also accounting for semantic risk and dynamic replanning. The important result is that the additional cost does not reduce success rate, and it substantially lowers raw semantic exposure and zone occupancy.

6 Analysis

References

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